Change Log

Measurement of $\nu_\mu + \bar{\nu}_\mu$ Charged-Current Single Pion Production on Argon in the NuMI Beam

Corrections that changed a result

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Abstract

This document records every correction that materially moved a number in the CC$\pi^\pm$ NuMI analysis, with the value before and after. It exists so that a reader of the analysis note never has to work out whether a given table predates a given fix, and so that a result quoted from an older draft can be identified as superseded rather than merely different.

Most entries are normalisation errors. That concentration is itself worth stating: the extraction divides by four numbers — the data exposure, the integrated flux, the argon target count, and the per-run Monte Carlo exposure — and eight separate defects came from one of those four differing between configurations without anything detecting it. The response is `norm_manifest.py`, which reads the authoritative configuration files and asserts that they agree, so that a future drift fails a check instead of producing a wrong cross section.

Two corrections changed physics conclusions rather than only normalisation, and the conclusions they invalidated are marked as withdrawn here and in the analysis note: the non-uniform-grid regularisation defect, which had collapsed several additional-smearing matrices towards rank one, and the beam-axis convention, under which reconstructed angular and transverse observables were not the same quantities as the generator predictions they were being compared against.

Contents

1 How to read this document

Each entry states what was wrong, how it was found, what changed numerically, and what — if anything — had to be withdrawn. "Before" values are quoted from the draft in which they appeared and should not be used; they are given so that an older table can be recognised.

Entries are ordered by the stage of the analysis they affect, not chronologically: exposure and flux first, then the Monte Carlo weighting, then the unfolding, then the systematic covariance.

2 Exposure, flux, and target normalisation

2.1 Obsolete integrated flux (factor $\sim$3.48)

An outdated flux file was used to convert the extracted event rate into a cross section. The integrated flux it supplied was larger than the correct value by a factor of approximately 3.48, so every cross section produced with it was smaller than the truth by the same factor.

Effect. All cross sections, all configurations. Any absolute normalisation from a draft predating this fix is wrong by $\sim$3.48 and must not be quoted. Shapes were unaffected, because the error was a single multiplicative constant.

2.2 Configuration-specific flux was not being used

The RHC and combined extractions were normalised with the FHC integrated flux. The three exposures see genuinely different spectra, and the correct integrated fluxes are

$$\Phi_{\mathrm{FHC}} = 6.81159 \times 10^{-10}, \quad \Phi_{\mathrm{RHC}} = 6.44646 \times 10^{-10}, \quad \Phi_{\mathrm{comb}} = 6.60865 \times 10^{-10}$$

in $\mathrm{cm}^{-2}\,\mathrm{POT}^{-1}$.

Effect. RHC cross sections rose by 5.7% and combined by 3.1%. FHC was already correct and did not move. The flux is now set by an explicit `Flux` line in each cross-section configuration file rather than inherited, and `norm_manifest.py` checks that all configurations of a given exposure agree on it.

2.3 Per-mode rather than per-run exposure weighting

Data and Monte Carlo exposures were pooled per horn mode and a single scale factor applied. Because the data-to-Monte-Carlo exposure ratio differs substantially between runs, this weighted the result by how much simulation happened to exist for each run rather than by the data actually collected. Each run is now scaled by its own $S_r = \mathrm{POT}_r^{\mathrm{data}}/\mathrm{POT}_r^{\mathrm{MC}}$.

Effect. Changed the relative weight of runs within every configuration. This is also the change that makes the framework ready for real data, where the per-run exposures are not free parameters.

2.4 Duplicated run-total POT in split files

Two conventions coexist in the processed samples. In `processed/` a run split across several files repeats the *run-total* exposure in every file; in `processed/w/` each file carries its own *true* exposure. Dividing by the per-file value broke the first convention and summing over files broke the second: the first made the inclusive tune come out a factor of two low, the second drove an RHC proton-tagged cross section negative.

The resolution is to normalise by the sum of *distinct* per-run exposures, which is correct under both conventions.

Effect. At the time of the fix the RHC $W_{\pi p}$ cross section moved from -0.194 to $+0.578$ (in units of $10^{-38}\,\mathrm{cm}^2/\mathrm{Ar}$), and the inclusive RHC p_μ result was unchanged at 0.860. Results retracted on the strength of the negative value were reinstated. Both numbers have since moved again under the central-value weighting fix below; the current values are in the analysis note.

3 Monte Carlo weighting

3.1 Central-value weight applied twice to the fake data

This is the largest single correction and it affected every result.

The fake data are generated externally by `throw_perrun_*.C`, which folds the tuned central value, the PPFX central value and the normalisation into the event multiplicity and then writes the per-event weights back as unity — the throw is a Poisson realisation of an already-weighted expectation. The framework's `build_universes` nevertheless substituted the *weighted* central-value histogram for the raw counts, on both the reconstructed and the truth side, so the central value was applied a second time.

How it was found. By asking directly whether the unfolded result was being compared against the input that generated it. Four independent checks confirmed the diagnosis: 1621.0 is a direct count of selected events across the four FHC files; 1721.9 equals both `FakeDataMC_0_reco` and `onBNB_reco` − `extBNB_reco`; run 1 gives 558.0 entries unweighted against 599.9 weighted for the same 558 events; and the throw's own expectation of 1567.1 agrees with the thrown count at 1.3σ.

Effect. The mean inclusive closure ratio moved from 1.106 to 1.001, and all 51 extracted cross sections fell by between 3.8% and 12.9%. Every absolute cross section from a draft predating this fix is high by roughly a tenth.

Withdrawn. Two earlier explanations of the closure excess — one attributing it to the unfolding covariance and Wiener filter, one to A_C smearing — were both wrong and are retracted. They are retained in the technical supplement as recorded dead ends rather than deleted.

3.2 Generator prediction histograms truncated at the overflow

Generator predictions were read without their overflow contents, so the highest-momentum region was silently missing from the comparison curves.

Effect. Model comparison only; the measured cross sections were unaffected. Any generator χ^2 from a draft predating the regeneration is invalid.

4 Unfolding

4.1 Second-derivative regularisation on a non-uniform grid

The Wiener-SVD second-derivative regulariser used the uniform-grid $(1, -2, 1)$ stencil on bins whose widths differ by factors of ten to fifteen. On such a grid that operator is not an approximation to the second derivative of anything, and its near-null direction is not the physically flat one. The Wiener filter can then retain that direction alone, which drives the additional-smearing matrix A_C towards rank one and destroys the shape information the measurement is supposed to carry.

The corrected operator differences the *density* on a non-uniform grid, reducing to the old stencil when the widths are equal, with one-sided first-derivative rows at the ends so that the operator stays non-singular without asserting that the distribution vanishes at the edges — which it does not, since $\cos\theta_\mu$ peaks at the upper edge.

Effect on the conditioning measure s_2/s_1 of A_C:

Observable	before	after
FHC θ_μ	0.015	0.51
combined θ_μ	0.045	0.38
proton-tagged $W_{\pi p}$	0.022	0.60
proton-tagged p_n	0.21	0.88

Withdrawn. The earlier conclusion that θ_μ "repairs" what $\cos\theta_\mu$ cannot is retracted: under the defective operator every model was being collapsed onto a common shape, so the apparent agreement was an artefact of the regularisation rather than a property of the data. After the fix θ_μ discriminates between models again and prefers the tune that generated the fake data.

This entry is the clearest illustration of why closure alone is not a sufficient validation. A rank-one extraction closes perfectly against its own correspondingly smeared truth while retaining almost no shape information.

4.2 Angular and transverse observables were defined about detector z

The NuMI neutrinos arrive approximately 28.6° away from the detector z axis, but reconstructed angles and transverse kinematic imbalance were being computed about z while the generator predictions were computed about the neutrino direction. These are not the same observables. Truth-level quantities now use the per-event neutrino direction and reconstruction uses the target-to-vertex direction; the two constructions differ only at the milliradian level.

Effect. 64.6% of signal events change $\cos\theta_\mu$ bin and 44.6% change $\cos\theta_\pi$ bin. The mean δp_T moves from 0.742 to 0.264 GeV/c. The muon and pion momenta, the opening angle $\theta_{\mu\pi}$, the hadronic mass $W_{\pi p}$, and the event selection are all unaffected.

Withdrawn. All proton-tagged transverse-kinematic model comparisons made before this correction are withdrawn, because the measured and predicted quantities were different observables.

4.3 Predictions were smeared by A_C twice

Every model prediction plotted or compared against the data was multiplied by the additional-smearing matrix twice: once by the extractor, which applies A_C to each entry of the prediction map in place, and again on the plotting path. Confirmed exactly, as $A_C^2 \cdot \text{raw} = \text{plotted} \times \text{width}$ to 1.0000 in every bin.

How it was found. By testing whether the released A_C lets an external user reproduce a published generator curve. It did not — the discrepancy was 31% on the integral and varied from 1.21 to 1.55 across bins, so it was not a normalisation — and the residual factor turned out to be A_C itself.

Effect. Model χ^2 values only. The anomaly the note previously attributed to A_C suppression of sharply peaked generators was this bug:

Observable	Model	before	after
FHC θ_μ	NEUT	40.60/5	8.96/5
FHC θ_μ	NuWro	38.62/5	9.43/5
FHC $\cos\theta_\mu$	NEUT	29.78/5	4.90/5
FHC $\cos\theta_\mu$	NuWro	29.10/5	4.93/5
FHC p_μ	GENIE	5.52/7	0.57/7

Not affected. The unfolded cross sections (FHC p_μ integrates to 0.8275 before and after) and the closure test, because the fake-data truth was the one curve not passed through the second multiplication. The previous table also predated the central-value weighting fix, so part of the shift in its "truth" column is that correction rather than this one.

Withdrawn. The argument for preferring θ over $\cos\theta$ on the grounds that the $\cos\theta_\mu$ χ^2 was an artefact. The effect that motivated it does not exist.

5 Systematic covariance

5.1 Combined detector-variation exposure double-counted

The combined configuration pooled detector-variation samples per horn mode, and the denominator used a single file's exposure rather than the summed Monte Carlo exposure over all files of that variation type, so the combined detector systematic was inflated.

Effect. The total detector systematic fell from 42.3% to 24.6%, and the combined result moved from 0.910 to 1.027 times its expected value. FHC and RHC were bit-identical before and after, which is the check that isolates the defect to the combination step.

5.2 Proton-tagged detector systematics were absent entirely

Of the fifty-one extractions, the thirty-three proton-tagged ones carried *no* detector systematic at all: the `detVar` column was `n/a` for every proton-tagged observable in every configuration, so their quoted uncertainties omitted a term worth 17–43% in the corresponding inclusive results.

The cause was a configuration gap rather than missing data. The proton-tagged detector variation samples had been processed on 2026-08-20/21 and carry the required branches, but nothing referenced them: the `_w` file-properties tables contained no `detVar` lines, and `ccpi1p_systcalc_numi.conf` had no `DV` block at all. Both were needed; adding either alone does nothing.

Effect. The thirty-three proton-tagged universes were rebuilt with the nine detector variation samples included and re-unfolded from them. Every extraction in the analysis now carries a detector term. The proton-tagged terms run 10.5%–64.4% with a median of 23.9%, against 16.9%–42.8% for the inclusive results.

Check. Per-mode detector-variation double counting has occurred in this analysis before, and the combined configuration — which carries two per-mode sets where the single-mode configurations carry one — is the one exposed to it. Double counting inflates the combined result. Of the eight observables (of eleven) whose combined value falls outside its own FHC–RHC range, six fall *below* both and only two above, which is the finite-sample averaging-down expected of statistically limited detector-variation samples rather than a recurrence.

6 Open items

Two things remain outstanding, and neither is a correction awaiting application: they are limits of the present exposure and of the blinding policy.

The differential $W_{\pi p}$ shape. Withdrawn, and not recoverable by any systematic treatment. At this exposure no binning satisfies both the migration and the statistics requirement simultaneously: a two-bin split near 1.15 GeV/c^2 meets the response criterion but leaves about 137 events in the first bin, which the Wiener filter then suppresses, while wider bins restore the statistics and fail the response. The proton-tagged sample supports an integrated cross section.

External reproduction of the A_C-smeared comparison. The released additional-smearing matrix is bit-identical to the operator the code applies, and the released covariances reproduce the quoted uncertainties, but the path from a raw generator file to a published curve has not been demonstrated end to end: a prediction is converted on load and transformed again on plotting, and applying A_C to a raw generator histogram does not reproduce that generator's plotted curve. The smeared curves are released so that comparisons against the models shown in the note need no such reconstruction, but a reader smearing a *new* model has no worked example to check against. The fix is reference code plus a numerical unit test reproducing one published curve bin by bin.

Everything that requires unblinding. No beam-on signal region has been opened, so the control-region checks in data, the background constraints, and any statement that a result agrees or disagrees with a generator remain untested against real data. Every number in the analysis note is a property of the injected sample and the extraction machinery.

A caveat that is not an open item but reads like one. The closure χ^2 values have p-values clustered near unity (median 0.96 over the fifty-one extractions) because the measurement and the truth it is compared against share their systematics, so those terms cancel in the difference while still inflating the covariance used to judge it. Sizing the uncertainty band properly would need a restricted covariance or an ensemble of throws. This is a limitation of what the closure test demonstrates, not an error in the values.

7 Superseded values that may still be encountered

Quantity	superseded value	status
Mean inclusive closure ratio	1.106	now 1.001
RHC $W_{\pi p}$ cross section	-0.194	sign was a POT-convention artefact
Combined detector systematic	42.3%	now 24.6%
Combined result / expected	0.910	now 1.027
FHC θ_μ A_C conditioning	0.015	now 0.51
Mean δp_T	0.742 GeV/c	now 0.264 GeV/c (beam frame)
Proton-tagged `detVar`	`n/a`	now 10.5–64.4% (median 23.9%)
Proton-tagged σ, all obs.	2026-08-18 values	superseded 2026-08-31